

Weidong Wang

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RESEARCH INTEREST

My research interest lies in topics related to *Software Engineering and Large Language Model System*, with a current focus on projects related to *LLM4SE, Code Agent, and Bug Detection*. I am currently working on developing agentic systems to advance bug detection and automated program repair.

EDUCATION

- **Nanjing University** Sept. 2022 - Jun. 2026
B.S. in Computer Science (Elite Class)
◦ GPA: 4.54/5.00 (top 10%)
Nanjing, China
- **University of Illinois Urbana-Champaign** Aug. 2026 - Jun. 2028 (expected)
incoming M.S. in Computer Science
Urbana, USA

PUBLICATIONS

- [1] **[ICML'25] EPIC: Efficient Position-Independent Caching for Serving Large Language Models**
Junhao Hu, Wenrui Huang, **Weidong Wang**, Haoyi Wang, Tiancheng Hu, Qin Zhang, Hao Feng, Xusheng Chen, Yizhou Shan, Tao Xie
- [2] **[ACL'25] RaaS: Reasoning-Aware Attention Sparsity for Efficient LLM Reasoning**
Junhao Hu, Wenrui Huang, **Weidong Wang**, Zhenwen Li, Tiancheng Hu, Zhixia Liu, Xusheng Chen, Tao Xie, Yizhou Shan
- [3] **[Preprint] AnyPoC: Universal Proof-of-Concept Test Generation for Scalable LLM-Based Bug Detection**
Zijie Zhao, Chenyuan Yang, **Weidong Wang**, Yihan Yang, Ziqi Zhang, Lingming Zhang

RESEARCH EXPERIENCE

- **iSE Group, UIUC** Sept. 2024 - Now
Advised by [Prof. Lingming Zhang](#), [Chenyuan Yang](#) and [Zijie Zhao](#) Urbana, IL, USA
 - Developed a source-agnostic multi-agent framework to automate the generation of executable Proof-of-Concept (PoC) code to validate textual bug reports through independent execution scrutiny and a project-specific evolving knowledge base. The project has successfully reported **92 new vulnerabilities** across major open-source ecosystems, with **27 PoCs** directly adopted as official regression tests.
 - * Role in the project: Engineered a scalable bug-scanning infrastructure to manage heterogeneous build/run environments and agentic scanning for 10 massive codebases and reported **120+** bugs. Our recent finding in OpenSSL has been officially assigned **CVE-2026-35188**.
 - Benchmarking the performance LLMs against hybrid (LLM + Static Analysis) approaches for similar bug patterns to evaluate capabilities for problem generalization based on large codebases.
 - * Role in the project: Constructing a specialized dataset of 54 Linux kernel patch groups with similar bug patterns spanning 15 distinct Common Weakness Enumeration (CWE) types.
- **LLMs Group, Peking University** Mar. 2024 - Sept. 2024
Advised by [Prof. Tao Xie](#), [Dr. Yizhou Shan](#) and [Junhao Hu](#) Beijing, China
 - We designed a position-independent context caching framework **EPIC** [1] that overcomes the prefix-matching limitation of conventional context caching. This method increased KV cache reuse by enabling non-prefix matching through a new algorithm *LegoLink*, achieving an **8x** reduction in TTFT(*Time-to-First-Token*) and a **7x** throughput improvement.
 - * Role in the project: Implemented changes to the attention layer to recompute key-value caches and new interface functions that work with the latest version of vLLM in about 800 lines of Python code.
 - To address the $O(N)$ memory complexity of KV Cache in long-chain reasoning, we proposed **RaaS** [2] that employs an LRU strategy to retain milestone tokens, reducing complexity to $O(L)$ ($L \ll N$) while preserving accuracy and efficiency.
 - * Role in the project: Implemented the computational kernels for various attention acceleration methods and built an end-to-end testing framework for performance evaluation.

HONORS AND AWARDS

- **People's Scholarship (First Class)** Dec. 2023
Nanjing University
- **Special Scholarship for Undergraduates in Basic Science (5/20)** Dec. 2023
Nanjing University

SKILLS

- **TOEFL Score**: 5.5 (Reading: 5.5, Listening: 6.0, Speaking: 5.5, Writing: 5.0)
- **Technical Expertise**: C/C++, Python, LaTeX, Rust